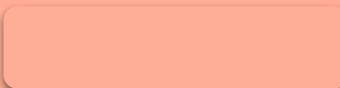


# Soulbound Tokens: Building a Decentralized Society

Based on the paper "Decentralized Society: Finding Web3's Soul" by Weyl, Ohlhaver, and Buterin

**Presenter:** Mahdi Mirzahosseini (s5436927)

**Date:** October 22, 2025



# The Web3 Paradox: centralization

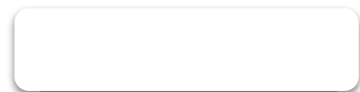
Web3 **can't** handle **who someone is** or **how they relate to others** in the ecosystem.

## Dependent on centralized Web2

OpenSea and Twitter to prove ownership and promotion

Social media profiles for Sybil resistance in DAOs

Coinbase or Binance for wallets



# The Solution: Souls & Soulbound Tokens (SBTs)

## **Soul = Wallet**

- permanent digital identity

## **Soulbound Token (SBT)**

- tokens held by the Soul.

## **Use cases of Soul and SBTs**

- Art & Verifiable Reputation
- Uncollateralized Lending
- Community Recovery
- Making DAOs More Robust



# Art & Verifiable Reputation

## Staking Reputation On-Chain

- **The Concept:** Artists and creators link digital items (NFTs, photos, videos) to their "Soul."
- **The Result:** This creates a verifiable, on-chain history or "provenance" for all of their work.

## Verifying Authenticity

- **For Art:** Buyers can inspect an artist's Soul to see their SBTs and confirm the work is legitimate.
- **For Media:** The same process helps identify deep fakes.

## How to verifying authenticity?

- **Trace:** Query the item's smart contract to find the creator's **Soul address**.
- **Inspect:** Use a blockchain explorer to view the **SBTs** held by that Soul.
- **Verify:** Check that the SBTs were issued by a **trusted source** (e.g., a university or guild) by confirming the issuer's public address.



# Soul Lending

**Web3:** Lacks trust for uncollateralized loans because assets are transferable and identities are not persistent.

**Traditional Finance:** Relies on centralized, opaque, and often discriminatory credit scores.

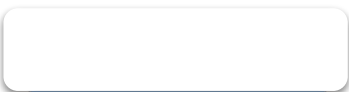
## The Soul Lending Solution

**Request:** A borrower requests a loan.

**Verification:** The lender reviews the borrower's Soul to assess their SBTs (work history, credentials, etc.).

**Issuance:** The loan is issued as a non-transferable "Loan SBT" that is publicly visible in the borrower's Soul.

**Repayment:** Once repaid, the Loan SBT is either burned or replaced by a "Proof of Repayment" SBT.



# Not Losing Your Soul

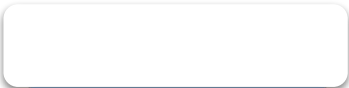
## The existing solution – “Social Recovery”

User appoints “guardians” who can change your keys by majority vote

## A More Robust Solution: Community Recovery

your communities become your recovery mechanism

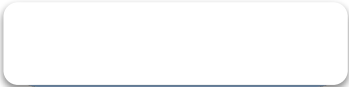
| Social Recovery                        | Community Recovery                                |
|----------------------------------------|---------------------------------------------------|
| <b>Fragile</b> as relationships change | <b>Robust</b> as security grows with your network |
| Needs <b>manual</b> updates            | Adapts <b>automatically</b>                       |
| <b>Vulnerable</b> to collusion         | <b>Resistant</b> to theft or collusion            |



# Making DAOs More Robust

Soulbound Tokens (SBTs) enable trustworthy, reputation-aware, and Sybil-resistant governance in DAOs

| Sybil Resistance                                                              | Fairer Voting                                             | Stronger Communities                                                     |
|-------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------|
| Differentiates unique Souls from bots by analyzing the constellation of SBTs. | Balances influence of correlated groups                   | Deters Vampire Attacks by rewarding loyal participants via Souldropping. |
| Assigns voting weight based on reputable SBTs                                 | Reduces the influence of special interests.               | Allows governance to adapt to the community's evolution.                 |
| <b>Outcome:</b> Ensures a more authentic "one Soul, one vote" system.         | <b>Outcome:</b> Encourages diverse support for proposals. | <b>Outcome:</b> Builds more resilient and engaged DAOs.                  |



# Decentralized Society" (DeSoc)

Cooperative, Plural, and Community-governed network

**The challenge is** finding the right balance between transparency and privacy

**deal with over-publicity**

store data off-chain

Possible storage options:

- Your own computer or phone,
- A trusted cloud (e.g., Google Drive),
- A decentralized system like IPFS (InterPlanetary File System).

**zero knowledge proofs & multi-party computation**

**designated-verifier proofs & Verifiable Delay Functions**

once you share data, it can leak.

**Negative reputation (Unirep-style)**

# SOUL BIRTH

What comes first: SBTs or community recovery?

## Proto SBTs

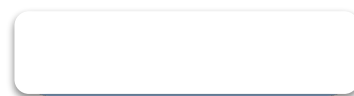
- revocable, transferable tokens

## Community Recovery Wallets

- Proof of Personhood (PoP)
- Custodial wallets
- Non-custodial wallets

## Proto-Souls

Norms can also shepherd Souls into existence



# **Thank you for you attention**

**I truly appreciate your time and  
consideration, especially during such a  
busy period.**

